

Vijay Kumar

Software Engineer | Java Backend & Real-Time Systems

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PROFESSIONAL SUMMARY

Backend engineer with production experience building real-time distributed systems using Java 17 and Spring Boot 3. Designed and shipped Nexus — a live multiplayer game platform with STOMP WebSocket sessions, PostgreSQL persistence, bcrypt authentication, and automated CI/CD via GitHub Actions. Strong foundation in concurrency, system design, and clean architecture. Targeting remote backend engineering roles.

TECHNICAL SKILLS

Languages	Java 17, JavaScript (ES2022), SQL, HTML / CSS Spring Boot 3, Spring Data JPA,
Backend	WebSocket (STOMP/SockJS), RESTful APIs, Maven PostgreSQL (Neon), H2 (testing),
Databases	JPQL bulk queries, Hibernate ORM Docker, GitHub Actions (CI/CD), Linux (Ubuntu 22.04),
Infrastructure	Vercel, Render, Oracle Cloud VM LLD / HLD, OOP, Multithreading & Concurrency, Event-
Architecture	Driven Design Data Structures & Algorithms, Operating Systems, Computer Networks
Fundamentals	

TECHNICAL PROJECTS

Nexus — Real-Time Multiplayer Game Platform

Java 17 · Spring Boot 3 · WebSocket · PostgreSQL

nexusgame.space | github.com/VijayKumarCode/nexus

- › Designed and deployed a full-stack real-time multiplayer game platform; currently live with concurrent user sessions managed via STOMP WebSocket and SockJS.
- › Built a stateful game engine using in-memory ConcurrentHashMaps for toss, challenge, and move state; added DB-backed recovery so active games survive server restarts.
- › Diagnosed Railway port-587 SMTP blocking through production log analysis; replaced JavaMailSender with Resend HTTP API (port 443), reducing activation email latency from 2-minute timeout to under 1 second.
- › Fixed JPQL enum syntax failures (User\$UserStatus inner-class reference) that caused scheduled tasks to crash every 10 seconds, silently breaking active games.
- › Secured the API by applying @JsonIgnore on password hash, activation token, and email fields; closed XSS surface in the lobby by replacing inline onclick username interpolation with data-attribute event delegation.
- › Configured GitHub Actions pipeline: builds JAR on push, copies via SCP to Oracle Cloud VM, and restarts the systemd service — zero-touch deployments on every commit.

TicTacToe_java — Network Multiplayer Engine

Java · Sockets · Multithreading

github.com/VijayKumarCode/TicTacToe_java

- › Built a socket-based two-player game server managing concurrent client sessions with dedicated threads per connection and synchronized shared game state — served as the architectural foundation for the Nexus platform.
- › Implemented server-side win detection, move validation, and real-time board synchronization across two connected clients.

DSA_Java — Algorithm & Data Structure Library

Java 17 · Algorithms

github.com/VijayKumarCode/Java-Data-Structures-and-Algorithms

- › Implemented and documented Trees, Graphs, Linked Lists, Hash Maps, and sorting algorithms in Java 17 with verified time and space complexity for each solution.

Portfolio Website

HTML · CSS · JavaScript · Vercel

[vijaykumarcodespace](https://vijaykumarcodespace.com) | github.com/VijayKumarCode/Vijay-Kumar-Portfolio

- › Built with vanilla HTML, CSS, and JavaScript — no frameworks. Deployed on Vercel with a custom domain and continuous deployment on every push to main.

EDUCATION

Master of Computer Applications (MCA)

AKGEC, Ghaziabad, Uttar Pradesh

Bachelor of Computer Applications (BCA)

LNMCBM, Muzaffarpur, Bihar